

Full data $X^n = (X_1, \dots, X_n)$, iid $\sim p \in \mathcal{P}$ (if H_0 true)

random / fixed split

$$D_0 = \{X_i : i \in I\}$$

$$D_1 = \{X_i : i \notin I\}$$

$$\hat{p}_0 = \arg \max_{p \in \mathcal{P}} \prod_{i \in D_0} p(X_i)$$

(null MLE on D_0)

pick $\hat{q}_1 \in \mathcal{Q}$ using D_1
(e.g. MLE / Bayes on D_1)

$$E = \prod_{i \in D_0} \frac{\hat{q}_1(X_i)}{\hat{p}_0(X_i)} \text{ evaluated on } D_0$$